

Fluctuations and Transient Events

VAFT Introductory Tutorial | Session 04

Why this session exists

- ▶ Session 03 handed you an equilibrium you can interrogate rather than assume.
- ▶ This one reads what will not hold still on top of it — and says how much of that reading you are *allowed* to claim.

A coil measures a derivative

$$S_{dB/dt}(f) = (2\pi f)^2 S_B(f)$$

An index fitted to raw pickup voltage is the field's **plus two**. VAFT never integrates behind an axis label.

An array sets the answer, not the algorithm

$$\varphi_{\text{phase}} = \varphi_0 - n\phi$$

- ▶ Two positions 120° apart $\rightarrow n$ modulo 3.
- ▶ Three positions 90° apart $\rightarrow n$ modulo 4.

Read the title before the number

- ▶ The fit states how many *distinct* toroidal positions answered, and which probes it left on another timebase.
- ▶ `available_plots` says a call will not raise — not that its answer is worth having.

Session workflow

timing → spectrogram → spectrum and band power → toroidal phase → rational surfaces

Where the chain stops

- ▶ No resolver from (m, n) to a surface; no rotation measurement, so no mode tracks; no event classifier of any kind.
- ▶ What you may write down: f and its drift, the band power against a measured floor, n modulo the array's alias step, ψ_N of each bare q surface — and that the pairing between them is undetermined.